

2025

PROJECT DESCRIPTION

Digital Waybill Management
System for Outrite Africa

PROJECT DURATION: 4 MONTHS

ROLE: FULL-STACK DEVELOPER

CLIENT: OUTRITE AFRICA (PTY) LTD - LOGISTICS COMPANY

STATUS: LIVE IN PRODUCTION



DHIALI DIGITAL DESIGNS



PROJECT OVERVIEW

Developed a comprehensive digital waybill management system for Outrite Africa, a logistics company, transforming their entirely paper-based operations into a modern, cloud-based solution. The system was built as a custom WordPress plugin and seamlessly integrated into their existing website hosted on domains.co.za, providing secure access to authorized users while maintaining complete data separation between company staff and external clients.

THE CHALLENGE

Outrite Africa's paper-based waybill system created significant operational bottlenecks:

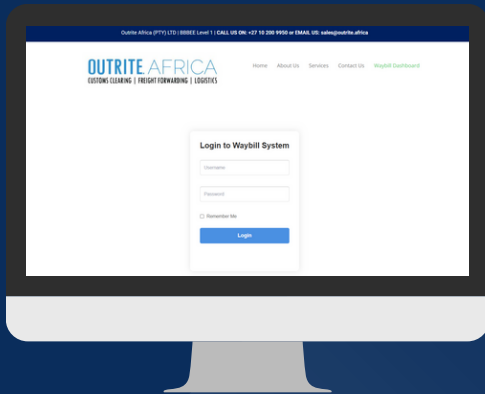
- Manual data entry consuming valuable staff time
- Frequent errors in volume mass and weight calculations
- Lost proof of delivery documents
- Overdue shipments going unnoticed
- No self-service access for external clients

THE SOLUTION

A production-ready WordPress plugin that completely eliminated paper-based processes while improving accuracy and efficiency across all logistics operations. The system automates previously manual tasks like generating unique waybill numbers, calculating volume mass and weights and formatting professional PDFs while adding capabilities that were impossible with paper such as real-time overdue tracking, instant Excel exports and 24/7 client access to shipment data. Built with robust security measures including role-based permissions, SQL injection prevention and file validation the platform supports unlimited concurrent users without number conflicts or data loss. The solution integrates Google Maps API for intelligent address entry and features an intuitive interface that requires minimal training, ensuring smooth adoption across users with varying technical abilities.

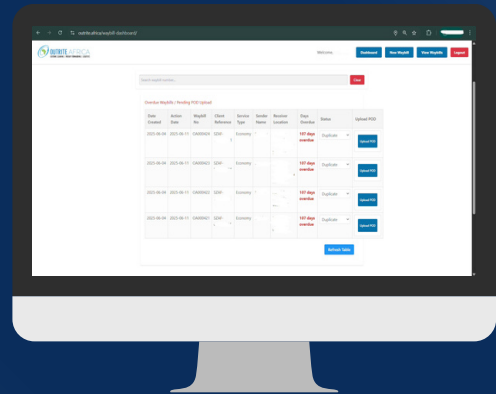
- 30+ Data fields per waybill
- 15+ Custom AJAX handlers
- 5 Delivery status stages
- 10MB Max file upload size
- 0 Waybill number conflicts

CORE FEATURES IMPLEMENTED



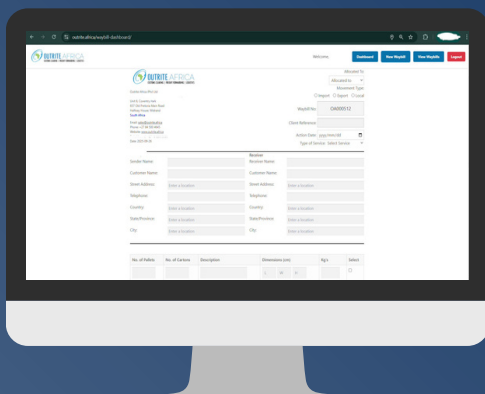
User Management & Security

- Role-based access: Admin (full system) vs. Client (own waybills only)
- Admin dashboard for credential generation, session management with "Remember Me"
- Password hashing, CSRF protection, SQL injection prevention, secure file storage



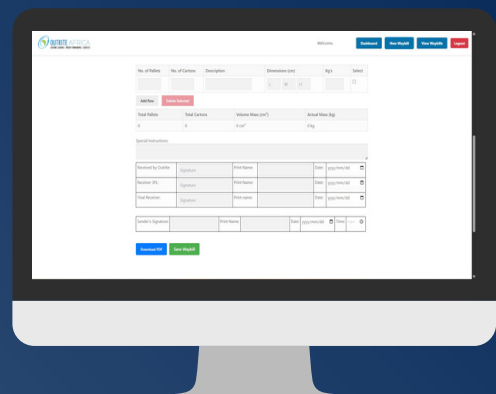
Intelligent Dashboard

- Overdue alerts with color-coded urgency (yellow: 1-4 days | red: 5+ days)
- Real-time waybill search, auto-refresh (5min), one-click POD upload
- Days overdue counter, direct invoice editing



Smart Waybill Creation

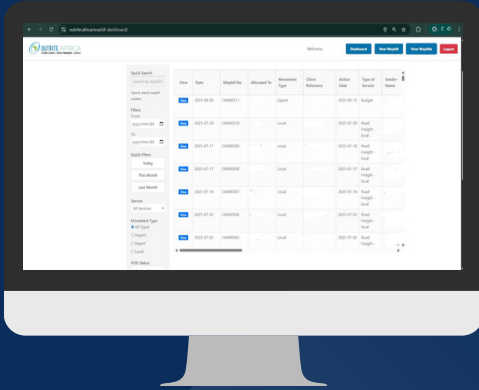
- Sequential OA##### numbering (never skips/duplicates with concurrent users)
- Google Maps API: address autocomplete, auto-extracts city/state/country, geolocation bias
- Pre-populated branding
- Movement type (Import/Export/Local), 8 service type options



Automated Calculations

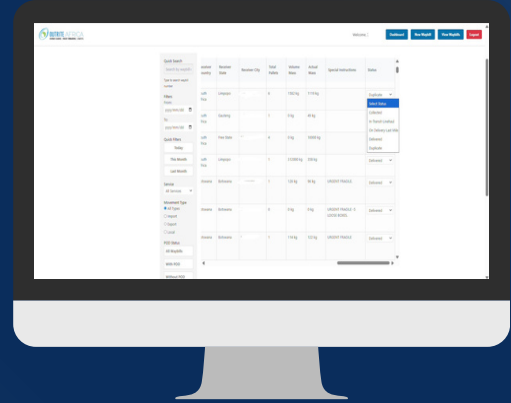
- Volume mass: $(L \times W \times H) / 5000$, dynamic unlimited rows with bulk delete, Real-time totals: pallets, cartons, mass
- Precise decimals (no rounding), instant updates, automatic recalculation
- Client-side jsPDF: instant downloads, branded templates, multi-page support
- Includes: full waybill data, goods tables, totals, signature sections (Sender/Receiver/3PL/Final)

CORE FEATURES IMPLEMENTED



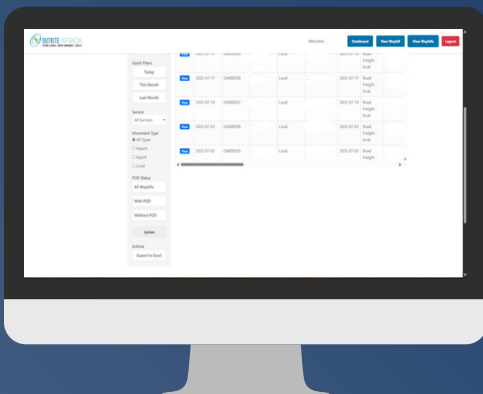
Advanced Filtering & Search

- Multi-criteria: date ranges, quick filters (Today/Month), service/movement types, POD status
- Instant live search, sticky headers, sortable columns, combined filter logic



Delivery Status & Workflow

- 5-stage tracking: Collected → In-Transit → On Delivery → Delivered → Duplicate
- Admin-only editing, visual indicators, real-time system-wide updates



Excel Export

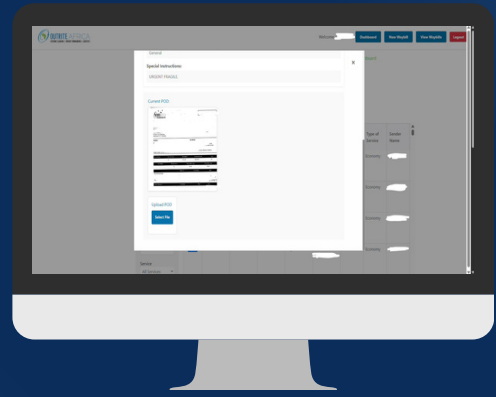
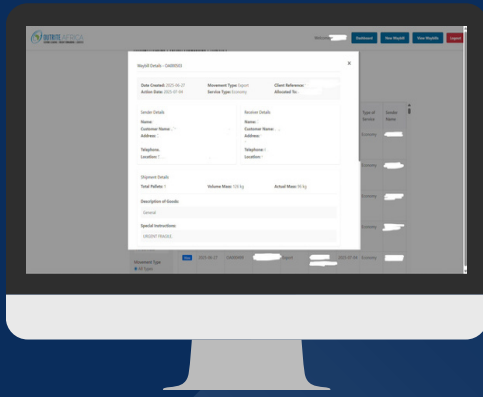
- One-click CSV export with UTF-8 encoding, applies all active filters
- All fields included: dates, references, sender/receiver data, goods, dimensions, weights, invoices
- Chronological sorting, dated filenames



Performance & Architecture

- AJAX-powered (no page reloads), efficient indexed database queries
- Client-side calculations, conditional script loading, lazy loading modals
- Database transactions with rollback, handles unlimited waybills + concurrent users

CORE FEATURES IMPLEMENTED



Waybill Viewer & POD Management

- Detailed table view, interactive modal popup with complete waybill details
- POD upload: JPEG/PNG/PDF (10MB max), image preview, auto-removes from overdue list
- Upload tracking, secure storage, old file cleanup

TECHNICAL IMPLEMENTATION

Backend Architecture

Platform: WordPress Plugin Architecture

Database: MySQL with 3 custom tables:

- wp_waybill_users - Authentication & permissions
- wp_waybills - 30+ fields per waybill, indexed on waybill_number & dates

Security:

- Prepared SQL statements (injection prevention)
- WordPress nonce verification (CSRF protection)
- Session validation, file upload validation, role-based access control
- Database transactions with rollback on errors
- Comprehensive PHP error logging

Frontend Development

Technologies: HTML5, CSS3, JavaScript (jQuery)

AJAX Architecture: 15+ custom handlers

- No page reloads, real-time validation, async file uploads
- Dynamic table updates, progress indicators

UI/UX Features:

- Toast notifications, loading spinners, form state management
- Smooth animations, responsive mobile-friendly design

API Integrations

Google Maps Places API:

- Address autocomplete, component extraction (street/city/state/country)
- Geolocation bias, error handling

jsPDF + jsPDF-AutoTable:

- Client-side PDF generation, automated table formatting

File & Data Management

POD Storage: WordPress Media Library, custom /pod-images/ folder

- Duplicate prevention, old file cleanup, .htaccess protection

Data Processing:

- Client-side calculations for instant feedback
- Multi-row form serialization, location data parsing
- Timezone-aware timestamps, UTF-8 BOM CSV export



KEY ACHIEVEMENTS



Business Impact

- Eliminated paper-based operations: Fully digitized waybill creation, tracking and POD management
- Improved efficiency: Automated calculations and data entry reducing creation time
- Enhanced accountability: Complete audit trail with timestamps and user tracking
- 24/7 client access: External clients view shipment data anytime without phone calls
- Zero calculation errors: Automated volume mass and totals eliminate manual mistakes
- Instant reporting: Real-time Excel exports for business intelligence and analysis



Technical Achievements

- Zero downtime integration: Deployed to live WordPress site without disrupting operations
- Scalable architecture: Handles unlimited waybills with efficient database indexing
- Concurrent user support: Multiple users create waybills simultaneously without number conflicts
- Cross-browser compatible: Tested on Chrome, Firefox, Safari, Edge
- Performance optimized: Lazy loading, conditional scripts, optimized queries
- Maintainable codebase: Well-documented modular PHP and JavaScript architecture



Security Implementation

- Multi-layer protection: WordPress authentication, session control, CSRF protection (nonces), SQL injection prevention, file validation, XSS sanitization
- Data privacy: Strict company/client data separation with role-based filtering
- Secure storage: Protected upload directories with .htaccess access restrictions

TECHNOLOGIES USED

Backend:

- PHP 7.4+
- WordPress 5.8+ (Custom Plugin Development)
- MySQL 5.7+
- WordPress REST API
- WordPress Session Management

Frontend:

- JavaScript (ES6+)
- jQuery 3.x
- HTML5
- CSS3 (Custom Styling with Animations)

APIs & Libraries:

- Google Maps Places API
- jsPDF 2.5.1
- jsPDF-AutoTable 3.5.28

Development Tools:

- WordPress Plugin API
- AJAX (Asynchronous JavaScript and XML)
- JSON for data exchange
- FormData API for file uploads

Hosting & Infrastructure:

- domains.co.za hosting platform
- WordPress hosting environment
- SSL/HTTPS security

CHALLENGES OVERCOME

1. **Sequential Numbering with Concurrency:** Implemented database-level unique constraints preventing duplicate waybill numbers when multiple users create waybills simultaneously.
2. **Large File Uploads:** Configured WordPress media settings and chunked upload handling for POD documents up to 10MB.
3. **Complex Calculations:** Built robust JavaScript engine handling edge cases (empty values, zero dimensions, special characters) while maintaining decimal precision.
4. **Role-Based Data Filtering:** Designed dynamic SQL queries with conditional WHERE clauses filtering by user permissions without code duplication.
5. **Google Maps API Integration:** Handled API rate limiting, network failures and missing data with graceful fallbacks and user-friendly error messages.
6. **PDF Generation:** Optimized jsPDF for variable-length content, multi-page documents and text wrapping while maintaining professional formatting.
7. **Real-time Table Updates:** Implemented efficient AJAX refresh updating only changed rows instead of reloading entire tables.
8. **Cross-browser Compatibility:** Resolved file upload API, date input and CSS grid issues across Chrome, Firefox, Safari and Edge.

FUTURE ENHANCEMENTS (POTENTIAL)

- SMS/Email notifications for status changes
- Barcode/QR code generation for waybill tracking
- Mobile app for drivers with offline capability
- Automated invoice generation based on completed waybills
- Analytics dashboard with charts and KPIs
- Integration with third-party shipping APIs
- Digital signature capture for proof of delivery
- Multi-language support for international operations

Successfully delivered a production-ready system that transformed Outrite Africa's operations from paper-based to fully digital. The system is currently in active use, handling all company waybill operations with positive feedback from both internal staff and external clients. The solution has improved operational efficiency, reduced errors and provided management with valuable insights through data analysis capabilities.